Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

LESSON PLAN

For

AI Vibe Coding for Multi-Agents System

TGS Ref No: TGS-2020503207

Conducted by

Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

Version 1.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	11 August 2026	Initial release — 2-day lesson plan for AI Vibe Coding for Multi-Agents System, aligned to the published course outline (Topics 1-5) and the 11 hands-on labs.	Dr Alfred Ang
1.1	11 August 2026	Labs restructured from single Markdown files into one folder per lab, each holding a runnable Python script and a Google Colab notebook; corrected the assessment durations to the papers on file (WA 50 minutes, PP 75 minutes) and the daily timing to 9:30am-7:00pm.	Dr Alfred Ang

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Course Information

Course Title	AI Vibe Coding for Multi-Agents System
WSQ Course Reference	TGS-2020503207
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Skills Framework	Analytics and Computational Modelling (ICT-DIT-3001-1.1)
Duration	2 days · 8 instructional hours per day (16 hours), plus assessment (WA 50 minutes + PP 75 minutes)
Daily Timing	9:30 am – 7:00 pm — 8 instructional hours per day. Breaks are additional and are not counted as instructional time: a 1-hour lunch and two 15-minute tea breaks (570 minutes elapsed, 480 instructional).
Mode	Instructor-led, hands-on multi-agent development labs in Python
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain the components of a modern AI agent — skills, memory, tools and the Model Context Protocol (MCP) — and the progression from single-agent to multi-agent collaboration.
- LO2: Apply a structured vibe coding workflow with context engineering to specify, generate and verify agent code reliably.
- LO3: Build a collaborative multi-agent system with the OpenAI Agents SDK using structured outputs, tool calling and supervisor routing, and deploy it with Streamlit.
- LO4: Build a collaborative multi-agent system with the Google Gemini Agent SDK and deploy it with Streamlit.
- LO5: Orchestrate tools through MCP and design hierarchical sub-agent architectures for delegated, specialised work.

Skills Framework Alignment

TSC Title: Analytics and Computational Modelling | TSC Code: ICT-DIT-3001-1.1

- A1 - Identify appropriate statistical algorithms and data models to test hypotheses or theories
- A2 - Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements
- A3 - Utilise a range of statistical methods and analytics approaches to analyse data
- A4 - Evaluate the performance and suitability of models against business requirements

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 50 minutes, open book.
- Practical Performance (PP) — hands-on multi-agent build tasks, 75 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The final assessment is conducted at the end of Day 2.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Training Resources

- Course slide deck (Trainer Slides / Learner Slides) — downloadable from the LMS/TMS portal.
- Learner Guide — full detailed step-by-step instructions for all 11 hands-on labs.
- Lab code repository: https://github.com/tertiarycourses/TGS-2020503207-AI-Vibe-Coding-for-Multi-Agents-System
- Learner laptop with Python 3.11 or later, an IDE or terminal, and internet access.
- API keys for the model providers used in the labs (OpenAI and Google AI Studio).
- LMS/TMS portal: https://lms-tms.tertiaryinfotech.com

Course Schedule

Day 1 — Agent Foundations & Vibe Coding for Multi-Agent Systems

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, learning outcomes, course outline and mandatory digital attendance (AM)
10:00–11:00	60 min	**Topic 1 — Modern Agent Foundations: what a modern AI agent is, the reason-act-observe loop, agent skills, memory, tools and the Model Context Protocol; the progression from single-agent to multi-agent collaboration (concepts + demonstration)**
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Hands-on: Lab 1: Build Your First Tool-Calling Agent; Lab 2: Agent Memory and Reusable Agent Skills
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	**Topic 2 — Vibe Coding for Multi-Agent Systems: the structured vibe coding workflow, context engineering, the vibe coding tool landscape and agent**

		skills integration (concepts + demonstration). Digital attendance (PM)
15:00–16:30	90 min	Hands-on: Lab 3: Vibe Code an Agent with a Structured Workflow; Lab 4: Context Engineering and a Reusable Coding Skill
16:30–16:45	15 min	Tea break
16:45–17:30	45 min	**Topic 3 — Multi-Agent System Development with OpenAI Agents SDK: agents, runners, function tools and Pydantic structured outputs (concepts + demonstration)**
17:30–18:45	75 min	Hands-on: Lab 5: First Agent with the OpenAI Agents SDK
18:45–19:00	15 min	Day 1 recap, Q&A and confirmation of digital attendance

Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch and two 15-minute tea breaks (9:30 am – 7:00 pm = 570 minutes elapsed).

Day 2 — Agent SDKs, MCP, Sub-Agents & Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–10:15	30 min	**Topic 3 continued — supervisor routing, sub-agent delegation, handoffs and guardrails (concepts + demonstration)**
10:15–11:15	60 min	Hands-on: Lab 6: Supervisor Routing and Sub-Agent Delegation
11:15–11:30	15 min	Tea break
11:30–12:30	60 min	Hands-on: Lab 7: Deploy the Multi-Agent System with Streamlit
12:30–13:00	30 min	**Topic 4 — Multi-Agent System Development with Gemini Agent SDK: agent design in the Google ADK and comparison with the OpenAI Agents SDK (concepts + demonstration)**
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Hands-on: Lab 8: Build a Multi-Agent System with the Gemini Agent SDK; Lab 9: Deploy the Gemini Multi-Agent System with Streamlit. Digital attendance (PM)
15:30–15:45	15 min	Tea break
15:45–16:15	30 min	**Topic 5 — MCP and Sub-Agents: MCP servers and tool orchestration in depth,**

		and hierarchical sub-agent architecture (concepts + demonstration)
16:15–18:00	105 min	Hands-on: Lab 10: Build and Connect an MCP Server; Lab 11: Hierarchical Sub-Agents and Context Isolation
18:00–18:30	30 min	Course revision and summary, course feedback and TRAQOM survey
18:30–18:45	15 min	**Briefing for Assessment and Assessment digital attendance**
18:45–19:00	15 min	**Final Assessment administration — the Written Assessment (WA, SAQ, 50 minutes) and Practical Performance (PP, 75 minutes) are conducted as scheduled by the assessment centre**

Total instructional time: 480 minutes (8 hours), excluding the 1-hour lunch and two 15-minute tea breaks (9:30 am – 7:00 pm = 570 minutes elapsed).

Lab Reference (aligned to the course outline)

Topic	Weighting	Labs
Topic 01: Modern Agent Foundations	20%	Lab 1, Lab 2
Topic 02: Vibe Coding for Multi-Agent Systems	20%	Lab 3, Lab 4
Topic 03: Multi-Agent System Development with OpenAI Agents SDK	25%	Lab 5, Lab 6, Lab 7
Topic 04: Multi-Agent System Development with Gemini Agent SDK	20%	Lab 8, Lab 9
Topic 05: MCP and Sub-Agents	15%	Lab 10, Lab 11

Learning Outcome to Lab Mapping

Learning Outcome	Topic	Assessed by
LO1: Explain the components of a modern AI agent — skills, memory, tools and the Model Context Protocol (MCP) — and the progression from single-agent to multi-agent collaboration.	Topic 01 (Lab 1, Lab 2)	WA (SAQ) + PP
LO2: Apply a structured vibe coding workflow with context engineering to specify, generate and verify agent code reliably.	Topic 02 (Lab 3, Lab 4)	WA (SAQ) + PP

LO3: Build a collaborative multi-agent system with the OpenAI Agents SDK using structured outputs, tool calling and supervisor routing, and deploy it with Streamlit.	Topic 03 (Lab 5, Lab 6, Lab 7)	WA (SAQ) + PP
LO4: Build a collaborative multi-agent system with the Google Gemini Agent SDK and deploy it with Streamlit.	Topic 04 (Lab 8, Lab 9)	WA (SAQ) + PP
LO5: Orchestrate tools through MCP and design hierarchical sub-agent architectures for delegated, specialised work.	Topic 05 (Lab 10, Lab 11)	WA (SAQ) + PP